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A smoothing method for solving quadratic convex separable knapsack problems

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ABSTRACT
In this paper, we propose a gradient-type method for solving a quadratic separable knapsack problem, which is simpler than the existing methods and competitive in practice. The algorithm uses a smooth approximation of the dual function of the knapsack problem. Partial convergence and complexity results of the algorithm are presented. Moreover, to demonstrate its efficiency, illustrative computational results are reported.

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1. Introduction

In this work, we consider an approximation scheme to solve the following continuous quadratic knapsack problem:

$$\begin{cases} \min\limits_{x} & \frac{1}{2}x^T P x - a^T x \\ \text{s.t.} & b^T x = c \\ & \ell \leq x \leq u \end{cases} \tag{1}$$

where $x \in \mathbb{R}^n$ is the decision variable and $P = \text{diag}(p)$ is a diagonal matrix in $\mathbb{R}^{n\times n}$, with $p = (p_1, p_2, \ldots, p_n)$ and positive entries $p_i > 0$ for all $i = 1, \ldots, n$. The vectors a, b, ℓ and u belong to $\mathbb{R}^n$, with real entries a_i, b_i, ℓ_i and u_i, respectively, where $b_i \neq 0$ and $\ell_i < u_i$ for all $i = 1, 2, \ldots, n$, and c is a positive scalar.

The search for a solution to Problem (1) is attractive due to its applicability in several areas, such as resource allocation problems [4,7], Lagrangian relaxation through nondifferentiable optimization methods, including subgradient methods [13], stratified sampling [6,16], inventory systems [17] and manufacturing queuing networks [5].

Many works exist on how to solve Problem (1). Following Bretthauer [8], from the optimality conditions of (1), we can transform this problem into a one-dimensional problem,

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which consists of finding a root of a piecewise continuous and linear function. In [9,14], the authors present algorithms based on the search for breakpoints, while in [5,15,21], they propose algorithms based on fixing variables. Dai and Fletcher [12] proposed a two-phase algorithm, consisting of bracketing and secant phases. Recently, Cominetti et al. [11] proposed a Newton–Secant method to solve Problem (1), where they exploit the specific structure of the continuous quadratic knapsack problem to develop an efficient algorithm. The method converges in $O(n)$ iterations and has a total arithmetic complexity of $O(n^2)$.

In this work, we propose an algorithm based on the gradient method with quadratic regularization to solve a modified version of Problem (1), using a smoothing of the projection operator onto a box. Problem (1) is transformed into the problem of finding the root of a function.

$$\begin{cases} g(\lambda) = 0, \\ \lambda \in \mathbb{R}, \end{cases} \tag{2}$$

where $g(\lambda) := \sum_{i=1}^{n} b_i x_i(\lambda) - c$, $x(\lambda) = (x_1(\lambda), x_2(\lambda), \ldots, x_n(\lambda))$ and $x_i(\lambda) = \text{proj}_{[\ell_i, u_i]}\left(\frac{\lambda b_i + a_i}{p_i}\right)$, for each $i = 1, 2, \ldots, n$. The function g resulting from this reformulation is piecewise linear and nondifferentiable. Details about $g(\lambda)$ are presented in Section 2.

The lack of differentiability makes it difficult to apply derivative-based methods; therefore, smoothing techniques have been widely explored in the literature. In [2], Bertsekas [2] addressed non-differentiable optimization constrained problems by means of smooth approximations of the objective function involving terms of the type $\max(0, f_i(x_i))$. Chen and Harker [10] proposed a class of approximations for complementarity problems, in which the function $\max(0, \cdot)$ also appears in the natural residual of projection operator. Venceslau [20] employed the hyperbolic smoothing technique for solving optimization problems and, in particular cases, demonstrated its equivalence with ϵ-subgradients.

Inspired by the non-smooth Newton method proposed by Cominetti et al. [11] where they use lateral derivatives for solving problem (2), we develop a gradient-like method that employs real-function smoothing strategies, such as those mentioned above. Our approach shares similarities with that of Cominetti et al. [11], but with one essential distinction: the proposed algorithm does not require the computation of derivatives during its execution. Moreover, we prove that the method attains an approximate solution in at most $\mathcal{O}(\epsilon^{-2})$ iterations.

This paper is organized as follows: Section 2 presents the optimality conditions of Problem (1), which are essential for understanding the context of our proposal. In Section 3, we discuss some smoothing results. Section 4 introduces the algorithm together with convergence results. Section 5 reports numerical experiments that illustrate the promising performance of our method. Finally, Section 6 provides concluding remarks.

2. Preliminaries

Following [8] and references therein, we define the Lagrangian function of the problem (1) using $L : \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n_+ \times \mathbb{R}^n_+ \to \mathbb{R}$, where $\lambda \in \mathbb{R}$ is the equality constraint multiplier of $c - b^T x = 0$, $\mu,\ \omega \in \mathbb{R}^n_+$ are the multipliers of inequality constraints $\ell - x \leq 0$ and $x - u \leq 0$, respectively. The KKT conditions for the search for a solution to Problem (1) is

given by

$$\sum_{i=1}^{n} b_i x_i = c, \tag{3}$$

$$\ell_i \leq x_i \leq u_i, \quad i = 1, 2, \ldots, n \tag{4}$$

$$x_i p_i - a_i - \lambda b_i - \mu_i + \omega_i, \quad i = 1, 2, \ldots, n \tag{5}$$

$$\mu_i(\ell_i - x_i) = 0, \quad i = 1, 2, \ldots, n \tag{6}$$

$$\omega_i(x_i - u_i) = 0, \quad i = 1, 2, \ldots, n \tag{7}$$

$$\mu_i \geq 0, \quad i = 1, 2, \ldots, n \tag{8}$$

$$\omega_i \geq 0, \quad i = 1, 2, \ldots, n. \tag{9}$$

We take $\overline{x}_i(\lambda) = \frac{\lambda b_i + a_i}{p_i}$ for all $i = 1, 2, \ldots, n$. For each $\lambda \in \mathbb{R}$, if $\overline{x}_i(\lambda) \leq \ell_i$, we assume $\mu_i = \ell_i p_i - \lambda b_i - a_i$ and $\omega_i = 0$. If $\ell_i < \overline{x}_i(\lambda) < u_i$, we assume $\mu_i = 0$ and $\omega_i = 0$. If $\overline{x}_i(\lambda) \geq u_i$, we assume $\mu_i = 0$ and $\omega_i = \lambda b_i + a_i - u_i p_i$. The following expression

$$x_i(\lambda) = \text{proj}_{[\ell_i, u_i]} \overline{x}_i(\lambda) \tag{10}$$

that is projection of $\overline{x}_i(\lambda) = \frac{\lambda b_i + a_i}{p_i}$ over interval $[\ell_i, u_i]$, for each $i = 1, 2, \ldots, n$. It is easy to show that (10) satisfy the conditions (4)–(9). Let $x : \mathbb{R} \to \mathbb{R}^n$ be a function defined as

$$x(\lambda) = (x_1(\lambda), x_2(\lambda), \ldots, x_n(\lambda)), \tag{11}$$

where each $x_i(\lambda)$ is given in (10). By (4), $x(\lambda)$ is a solution to Problem (1) if and only if the following equality holds:

$$b^\top x(\lambda) = c. \tag{12}$$

Hence, solving Problem (1) is equivalent to finding a real value of λ that satisfies Equation (12), provided such a value exists.

Now let us define the function $g : \mathbb{R} \to \mathbb{R}$, given by

$$g(\lambda) := b^T x(\lambda) - c = \sum_{i=1}^{n} b_i x_i(\lambda) - c, \tag{13}$$

where $x(\lambda)$ is given by Equation (11). The function g is continuous, piecewise linear and non-decreasing. It may be non-differentiable at the breakpoints $\frac{p_i \ell_i - a_i}{b_i}$ and $\frac{p_i u_i - a_i}{b_i}$.

We have that $\lambda^* \in \mathbb{R}$ is a root of Equation (13) if and only if $b^\top x(\lambda^*) = c$. In other words, λ^* is a root of g if and only if $x(\lambda^*)$ is a solution to Problem (1). Therefore, the original minimization problem can be reformulated as the following root-finding problem:

$$\begin{cases} g(\lambda) = 0, \\ \lambda \in \mathbb{R}. \end{cases} \tag{14}$$

The nondifferentiability of g makes it challenging to use methods that need the derivative to solve Problem (14). We will look for smooth approximations of (13) to overcome this difficulty.

First, we give a reformulation of the projection operator on a box. Let us observe that

$$\text{proj}_{[\ell,u]}x = \max(\ell, \min(x, u)).$$

After some algebraic manipulations, we conclude that

$$\text{proj}_{[\ell,u]}x = \ell + \max(0, x - \ell - \max(0, x - u)). \tag{15}$$

Defining $r(\lambda)$ by $\max(0, \lambda)$, we rewrite (10) by

$$x_i(\lambda) = \ell_i + r\big(\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i)\big). \tag{16}$$

Replacing the right side (16) into (13), we can obtain

$$g(\lambda) = \sum_{i=1}^{n} b_i\Big(\ell_i + r\big(\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i)\big)\Big) - c, \tag{17}$$

where

$$\overline{x}_i(\lambda) = \frac{\lambda b_i + a_i}{p_i}$$

for all $i = 1, 2, \ldots, n$.

The next section is dedicated to the study of smooth approximations of the function $g(\lambda)$.

3. Smooth approximations of the function $g(\lambda)$

In this section, we consider a class of parametric smooth approximation functions or just approximation function of the function $r(\lambda) = \max(0, \lambda)$. This class of smooth functions are part of the smooth functions given in [3,10], and the properties will be essential for us to obtain a smooth approximation function of (17).

We obtain a smooth approximation of (17) by replacing $r(\lambda)$ with an approximation function $\phi(\lambda, \tau)$, as will be detailed in the remainder of this section.

Definition 3.1: Let $\phi : \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ be a smooth function. We say that $\phi(\lambda, \tau)$ is a smooth approximation with parameter τ or, simply, an approximation function of the function $r(\lambda)$, if there exists a continuous and strictly increasing function $\nu : \mathbb{R}_+ \to \mathbb{R}_+$ satisfying $\nu(0) = 0$ such that

$$|\phi(\lambda, \tau) - r(\lambda)| \leq \nu(\tau), \quad \forall\, \lambda \in \mathbb{R} \text{ and } \tau > 0. \tag{18}$$

As stated in Equation (18), the function $\phi(\lambda, \tau)$ provides a smooth approximation of $r(\lambda)$ that becomes increasingly accurate as τ approaches 0^+.

For the class of functions used to approximate the function $r(\lambda)$, we shall assume the following properties for the first and second derivatives of the approximation function $\phi(\lambda, \tau)$ with respect to λ:

(i) $\phi'(\lambda, \tau) : \mathbb{R} \times \mathbb{R}_+ \to [0, 1]$;

(ii) $\phi'(\lambda, \tau) \in C^1$ in relation to λ;
(iii) $\phi''(\lambda, \tau) > 0$ in relation to λ, i.e. $\phi'(\lambda, \tau)$ is strictly increasing for each $\tau > 0$;
(iv) $\lim_{\lambda \to -\infty} \phi'(\lambda, \tau) = 0$ and $\lim_{\lambda \to +\infty} \phi'(\lambda, \tau) = 1$.

Note that, by condition (iii), the function $\phi(\lambda, \tau)$ is strictly convex for each $\tau > 0$. It has been proven in [10] that if $\phi(\lambda, \tau)$ is an approximation of $r(\lambda)$ and is strictly convex, then

$$0 < \phi'(\lambda, \tau) < 1. \tag{19}$$

We present below two examples of approximation functions for $r(\lambda)$ that satisfy properties (i)–(iv):

Example 3.2: Based on [3], we provide the following examples:

(i) *Neural network smoothing function*:

$$\phi_1(\lambda, \tau) = \tau \ln(1 + e^{\frac{\lambda}{\tau}}), \quad \tau > 0,$$

where $v(\tau) = \tau \ln 2$.

(ii) *Interior point smoothing function*:

$$\phi_2(\lambda, \tau) = \frac{\lambda + \sqrt{\lambda^2 + \tau}}{2},$$

where $v(\tau) = \frac{\sqrt{\tau}}{2}$.

Remark 3.1: For the sake of simplicity, we chose these two approximation functions because they are known in the literature, already used in [3,10], for example, and fit naturally into our framework. They satisfy the required properties (i)–(iv) and provide a clear and practical way to illustrate the smoothing process, as we can see in Figures 1 and 2.

Consider the following example related to Problem (1):

Example 3.3: Let us examine the following quadratic knapsack problem:

$$\begin{cases} \min\limits_{x} & \frac{1}{2}(x_1^2 + x_2^2) - (x_1 - x_2) \\ \text{s.t.} & x_1 + 2x_2 = 10 \\ & 1 \le x_1 \le 10 \\ & 1 \le x_2 \le 5 \end{cases}$$

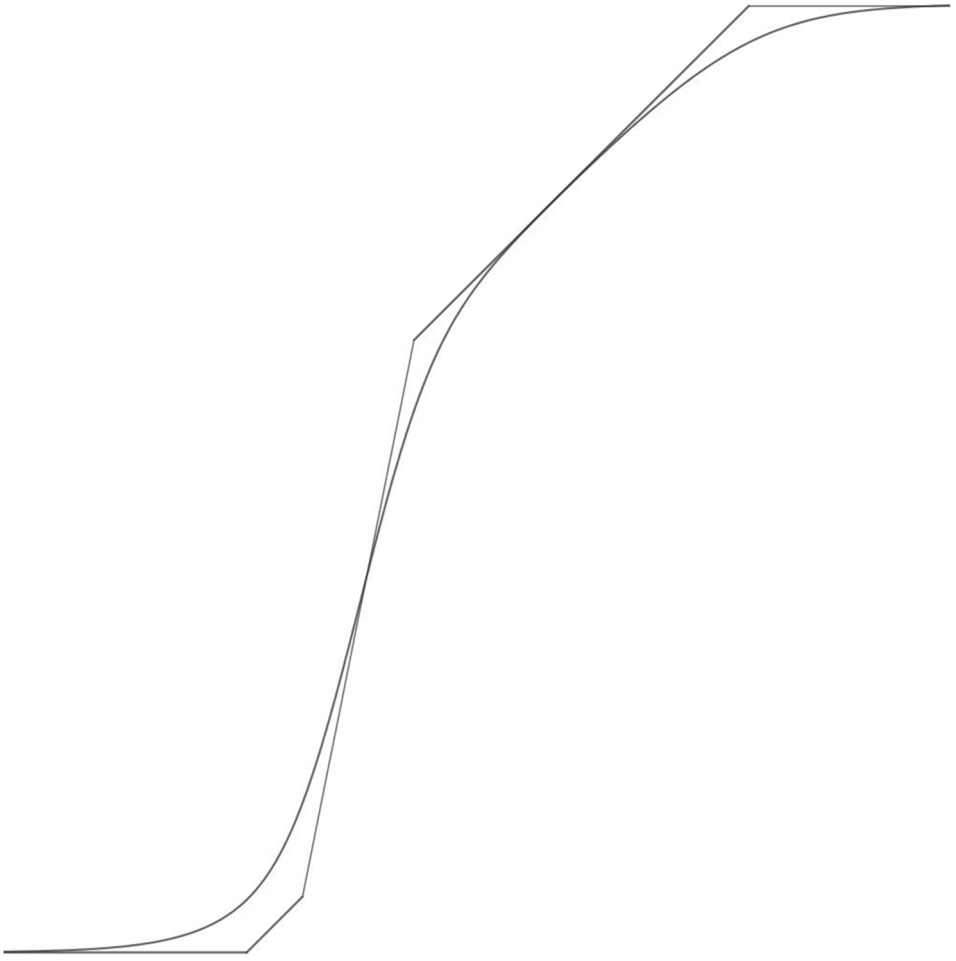

Figure 1. The graphs of the functions g, g_{ϕ_1} given in Example 3.3.

The function $g(\lambda)$ for the problem of Example 3.3 is given by

$$g(\lambda) = r(\lambda - r(\lambda - 9)) + 2r(2\lambda - 2 - r(2\lambda - 6)) - 7. \tag{20}$$

We define $\tau = 1$ and replacing the function $r(\cdot)$ in (20) by $\phi_1(\cdot, 1)$ and then by $\phi_2(\cdot, 1)$ we get $g_{\phi_1}(\lambda)$ and $g_{\phi_2}(\lambda)$, respectively, given by

$$g_{\phi_1}(\lambda) = \ln(1 + e^{\lambda - \ln(1+e^{\lambda-9})}) + 2\ln(1 + e^{2\lambda - 2 - \ln(1+e^{2\lambda-6})}) - 7$$

$$g_{\phi_2}(\lambda) = \frac{\lambda + 9 - \sqrt{(\lambda-9)^2+1} + \sqrt{(\lambda + 9 - \sqrt{(\lambda-9)^2+1})^2 + 4}}{4}$$
$$+ \frac{2\lambda + 2 - \sqrt{(2\lambda-6)^2+1} + \sqrt{(2\lambda + 2 - \sqrt{(2\lambda-6)^2+1})^2 + 4}}{2} - 7$$

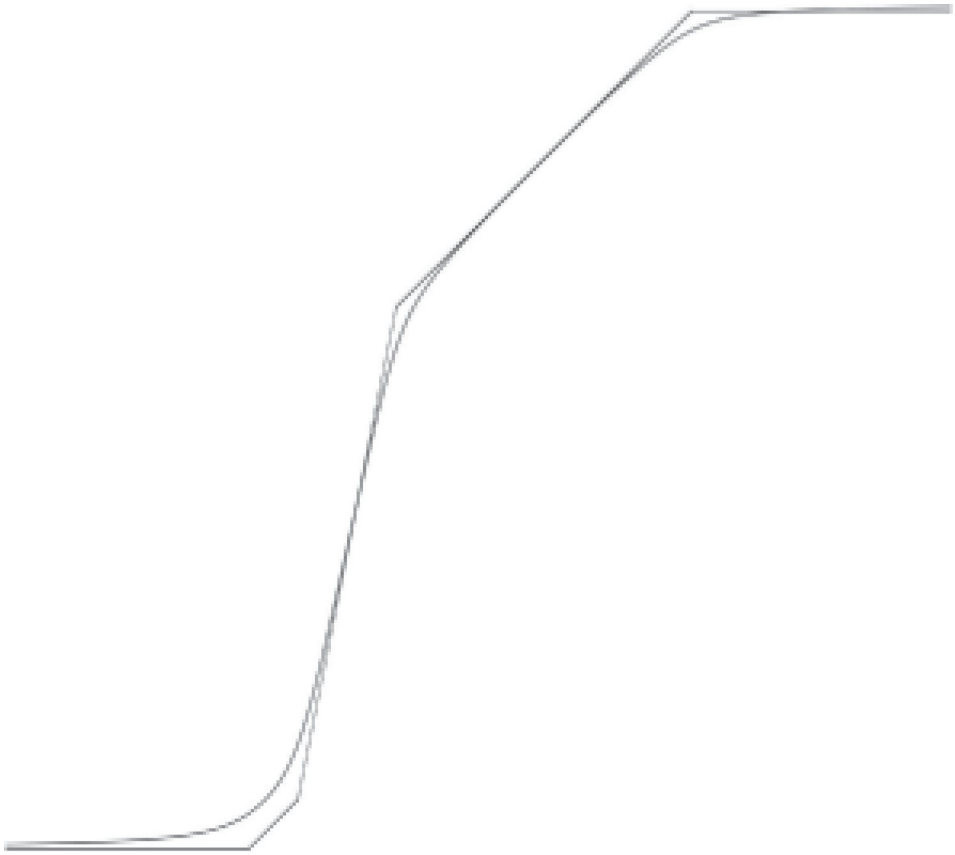

Figure 2. The graphs of the functions g, g_{ϕ_2} given in Example 3.3.

Figures 1 and 2 show how the graphs of functions $g_{\phi_1}(\lambda)$ and $g_{\phi_2}(\lambda)$ are close to the graph of function $g(\lambda)$, respectively.

In the following, we will replace the $\max(0, \lambda)$ by a parametric smooth approximation $\phi(\lambda, \tau)$ in (17), and we will show that the generated function is smooth and approximates to (17) when $\tau > 0$ approaches zero.

Replacing $x_i(\lambda)$ by $x_{i_\phi}(\lambda, \tau)$ in the function g given in (17), with

$$x_{i_\phi}(\lambda, \tau) = \ell_i + \phi\big(\overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau), \tau\big) \tag{21}$$

where $\phi(\cdot, \tau)$ is an approximation function of the function $r(\lambda)$. We get the following smooth function:

$$g_\phi(\lambda, \tau) = \sum_{i=1}^{n} b_i \left(\ell_i + \phi\big(\overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau), \tau\big)\right) - c. \tag{22}$$

The next results show that $g_\phi(\lambda, \tau)$ approaches $g(\lambda)$ as $\tau > 0$ approaches zero, thus ensuring that this is a smooth approximation of g.

Lemma 3.4: *Let α, $\beta \in \mathbb{R}$ be and $\tau > 0$. If $|\alpha - \beta| \leq \nu(\tau)$, with $\nu(\tau)$ according to Definition 3.1, then*

$$|\phi(\alpha, \tau) - \phi(\beta, \tau)| \leq 3\nu(\tau). \tag{23}$$

Proof: We have that

$$\begin{aligned} |\phi(\alpha, \tau) - \phi(\beta, \tau)| &= |\phi(\alpha, \tau) - r(\alpha) + r(\alpha) - r(\beta) + r(\beta) - \phi(\beta, \tau)| \\ &\leq |\phi(\alpha, \tau) - r(\alpha)| + |r(\alpha) - r(\beta)| + |r(\beta) - \phi(\beta, \tau)|. \end{aligned}$$

Since the function $r(t)$ is Lipschitz, with constant equal to 1 and by (18), immediately follows the inequality (23). ■

Lemma 3.5: *For all $\lambda \in \mathbb{R}$, $\tau > 0$ and for either $i = 1, 2, \ldots, n$, the following inequality holds:*

$$|x_{i_\phi}(\lambda,\ \tau) - x_i(\lambda)| \leq 4\nu(\tau) \tag{24}$$

Proof: By (18), follow that

$$|\overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau) - (\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i))| \leq \nu(\tau).$$

Assuming $\alpha = \overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau)$ and $\beta = \overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i)$ and using Lemma 3.4, we have that

$$|\phi(\overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau), \tau) - \phi(\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i), \tau)| \leq 3\nu(\tau).$$

Then

$$\begin{aligned} |x_{i_\phi}(\lambda,\ \tau) - x_i(\lambda)| &= |\phi(\overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau), \tau) \\ &\quad - r(\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i))| \\ &\leq |\phi(\overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau), \tau) \\ &\quad - \phi(\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i), \tau)| \\ &\quad + |\phi(\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i), \tau) - r(\overline{x}_i(\lambda) - \ell_i - r(\overline{x}_i(\lambda) - u_i))| \\ &\leq 4\nu(\tau). \end{aligned}$$

■

A consequence of (24) is that $\lim_{\tau \to 0^+} x_{i_\phi}(\lambda,$
$tau) = x_i(\lambda)$.

Proposition 3.6: *Let $\tau > 0$ be. There is a positive constant $C > 0$ such that, for all $\lambda \in \mathbb{R}$*

$$|g_\phi(\lambda,\ \tau) - g(\lambda)| \leq C \cdot \nu(\tau). \tag{25}$$

Proof: Follow that

$$|g_\phi(\lambda,\ \tau) - g(\lambda)| = \left|\sum_{i=1}^{n} b_i(x_{i_\phi}(\lambda,\ \tau) - x_i(\lambda))\right|$$
$$\leq \sum_{i=1}^{n} |b_i| \cdot |x_{i_\phi}(\lambda,\ \tau) - x_i(\lambda)|.$$

By Lemma 3.5 and then taking $C = 4\sum_{i=1}^{n} |b_i|$ follow the inequality (25). ■

Proposition 3.6 shows that the function $g_\phi(\cdot,\ \tau)$ is, in fact, an approximation of the function $g(\cdot)$. Furthermore, we will show that the error of this approximation can be controlled by the size of the variable τ, thus ensuring an accurate fit as τ approaches zero.

Lemma 3.7: *For all $\lambda \in \mathbb{R}$ and $\tau > 0$, we have*

$$0 < g'_\phi(\lambda,\ \tau) < \sum_{i=1}^{n} \frac{b_i^2}{p_i}. \tag{26}$$

Proof: Deriving the function $g_\phi(\lambda,\ \tau)$ with respect to λ, we have that

$$g'_\phi(\lambda,\ \tau) = \sum_{i=1}^{n} \frac{b_i^2}{p_i}\phi'(\overline{x}_i(\lambda) - \ell_i - \phi(\overline{x}_i(\lambda) - u_i, \tau), \tau)(1 - \phi'(\overline{x}_i(\lambda) - u_i, \tau)).$$

By (iii) and (iv), we have that $0 < \phi'(\lambda, \tau) < 1$ for all $\lambda \in \mathbb{R}$, it follows that (26). ■

In the following section, using $g_\phi(\lambda,\ \tau)$ with smooth approximation of g defined in (17), we propose a method to solve Problem (1).

4. A gradient-type method and its convergence results

In this section, we present the method proposed to solve Problem (1). Before introducing the method, we provide a theoretical analysis that justifies the choice of parameters and the adopted formulation.

4.1. Smoothing gradient-type algorithm

This section is dedicated to presenting a gradient-type method to solve Problem (1). The term Smoothing Gradient-Type Algorithm originates from Equation (27), introduced later in this section.

Since $g_\phi(\lambda,\ \tau)$ is differentiable at λ, then for each τ there exists a function $F_\tau(\lambda)$ such that $F'_\tau(\lambda) = g_\phi(\lambda,$
tau) and consequently $F''_\tau(\lambda) = g'_\phi(\lambda,\ \tau)$. By (26), we have that $F''_\tau(\lambda)$ is positive, thus F_τ is a convex function. Therefore, λ^* is a root of $F'_\tau(\lambda) = g_\phi(\lambda,\ \tau)$ if and only if it is a minimizer of F_τ.

We consider the following method with quadratic regularization:

$$\lambda_{k+1} \text{ is the solution of } \min_{\lambda} F_\tau(\lambda_k) + F'_\tau(\lambda_k)(\lambda - \lambda_k) + \frac{\rho}{2}(\lambda - \lambda_k)^2. \tag{27}$$

We observe that the function $\Phi(\lambda) = F_\tau(\lambda_k) + F'_\tau(\lambda_k)(\lambda - \lambda_k) + \frac{\rho}{2}(\lambda - \lambda_k)^2$ is a quadratic function defined on $\mathbb{R}$. Therefore, it has a unique minimizer, and consequently, Problem (27) has a unique solution λ_{k+1}.

It is easy to see that $\lambda_{k+1} = \lambda_k - \frac{F'_\tau(\lambda_k)}{\rho}$, that is, $\lambda_{k+1} = \lambda_k - \frac{g_\phi(\lambda_k, \tau)}{\rho}$.

Since $F''_\tau(\lambda) = g'_\phi(\lambda, \tau)$ is bounded above by $L = \sum_{i=1}^n \frac{b_i^2}{p_i}$, then $F'_\tau(\lambda)$ is Lipschitz with constant L. Thus

$$F_\tau(\lambda) \le F_\tau(\gamma) + F'_\tau(\gamma)(\lambda - \gamma) + \frac{L}{2}(\lambda - \gamma)^2, \tag{28}$$

see the proof of (28) in [22, Lemma 1].

Using $\lambda = \lambda_{k+1} = \lambda_k - \frac{F'_\tau(\lambda_k)}{\rho}$ and $\gamma = \lambda_k$ in (28), we obtain the following:

$$\begin{aligned} F_\tau(\lambda_{k+1}) &\le F_\tau(\lambda_k) + F'_\tau(\lambda_k)(\lambda_{k+1} - \lambda_k) + \frac{L}{2}(\lambda_{k+1} - \lambda_k)^2 \\ &= F_\tau(\lambda_k) + g_\phi(\lambda_k, \tau)(\lambda_{k+1} - \lambda_k) + \frac{L}{2}(\lambda_{k+1} - \lambda_k)^2 \\ &= F_\tau(\lambda_k) - \rho(\lambda_{k+1} - \lambda_k)^2 + \frac{L}{2}(\lambda_{k+1} - \lambda_k)^2 \\ &= F_\tau(\lambda_k) - \left(\rho - \frac{L}{2}\right)(\lambda_{k+1} - \lambda_k)^2, \end{aligned}$$

hence, if $\rho \ge L$, we obtain that $\frac{L}{2}(\lambda_{k+1} - \lambda_k)^2 \le F_\tau(\lambda_k) - F_\tau(\lambda_{k+1})$. Thus, as long as $\lambda_{k+1} - \lambda_k \ne 0$, the sequence $\{F_\tau(\lambda_k)\}$ is decreasing, and it is expected that the method converges. Hence, we assume that $\rho = L = \sum_{i=1}^n \frac{b_i^2}{p_i}$, that is,

$$\lambda_{k+1} = \lambda_k - \frac{g_\phi(\lambda_k, \tau)}{\sum_{i=1}^n \frac{b_i^2}{p_i}}.$$

Building on this analysis, we propose the following method to solve Problem (1).

Smoothing Gradient-Type Algorithm – SGTA

Step 0. Given $\lambda_0 \in \mathbb{R}$, $\epsilon > 0$, $\beta \in (0, 1)$ and $\tau_0 > 0$.

Step 1. Compute

$$\bar{\lambda}_{k+1} = \lambda_k - \frac{g_\phi(\lambda_k, \tau_k)}{\sum_{i=1}^n \frac{b_i^2}{p_i}} \tag{29}$$

and go to Step 2.

Step 2. If $|g(\bar{\lambda}_{k+1})| \le \epsilon$, stop and declare

$$x(\bar{\lambda}_{k+1}) = \big(x_1(\bar{\lambda}_{k+1}), x_2(\bar{\lambda}_{k+1}), \ldots, x_n(\bar{\lambda}_{k+1})\big)$$

as an approximation solution of Problem (1). Otherwise, go to Step 3.

Step 3. If $|g_\phi(\bar{\lambda}_{k+1}, \tau_k)| \leq \frac{\epsilon}{2}$, set $\tau_k := \beta\tau_k$, and go to Step 1. Otherwise, go to Step 4.
Step 4. If $|g_\phi(\bar{\lambda}_{k+1}, \tau_k)| > \frac{\epsilon}{2}$, set $\lambda_{k+1} = \bar{\lambda}_{k+1}$, $\tau_{k+1} = \tau_k$, $k := k+1$ and go to Step 1.

It is important to emphasize that, during the execution of Algorithm SGTA, it isnot necessary to use derivatives of the functions involved, which is advantageous, especially when the computation of these derivatives is prohibitive. Another point worth highlighting is that the parameter τ_k only needs to be updated until it reaches $\nu(\tau_k) \leq \frac{\epsilon}{2C}$, that is, from the moment this estimate is achieved, the parameters τ_k remain constant. Therefore, if the initial value τ_0 satisfies $\nu(\tau_0) \leq \dfrac{\epsilon}{2C}$, then all elements τ_k will remain constant and equal to τ_0; see details in Remark 4.1.

Remark 4.1: If τ_0 is chosen so that $\nu(\tau_0) \leq \frac{\epsilon}{2C}$, by (25), we have the following:

$$|g(\lambda)| \leq \nu(\tau_0) + |g_\phi(\lambda, \tau_0)| \leq \frac{\epsilon}{2} + |g_\phi(\lambda, \tau_0)|,$$

that is,

$$|g(\lambda)| - \frac{\epsilon}{2} \leq |g_\phi(\lambda, \tau_0)|.$$

Therefore, if $|g(\lambda)| > \epsilon$, we have $\frac{\epsilon}{2} \leq |g_\phi(\lambda, \tau_0)|$. This means that if τ_0 is chosen such that $\nu(\tau_0) \leq \frac{\epsilon}{2C}$, the parameter τ_0 does not need to be doubled, that is, $\tau_k = \tau_0$ for every iteration k.

4.2. Worst-case iteration complexity

In this section, we will show that Algorithm SGTA achieves the stopping criterion given in Step 2 in a finite number of iterations, where we analyse the worst-case iteration complexity of the proposed method.

The next lemma is useful to prove that the method reaches the stopping criterion established in Step 2 in a finite number of iterations, which is stated in Theorem 4.2.

Lemma 4.1: *For each k, let $F_{\tau_k}(\cdot)$ be a primitive of $g_\phi(\cdot, \tau_k)$. Then there exists an index k_0 such that the sequence $\{F_{\tau_{k_0}}(\lambda_k)\}$ is decreasing for all $k \geq k_0$ or the method stops up to iteration k_0.*

Proof: Since $\lim_{k\to+\infty} \tau_k = 0$, by continuity of ν and the fact that $\nu(0) = 0$, we have that $\lim_{k\to+\infty} \nu(\tau_k) = 0$. Thus, for the ϵ given in the method, there exists k_0 such that $\nu(\tau_k) \leq \frac{\epsilon}{2C}$ for all $k \geq k_0$.

We claim that τ_k does not need to be updated after iteration k_0. In fact, suppose the method does not halt at iteration $K > k_0$, then $|g(\lambda_k)| > \epsilon$. Hence, as $\nu(\tau_k) \leq \frac{\epsilon}{2C}$ and, by (25), we have

$$\epsilon < |g(\lambda_k)| \leq |g_\phi(\lambda_k, \tau_k)| + C\nu(\tau_k) \leq |g_\phi(\lambda_k, \tau_k)| + \frac{\epsilon}{2},$$

whence it follows that $|g_\phi(\lambda_k, \tau_k)| > \frac{\epsilon}{2}$. Therefore, τ_k does not need to be updated from iteration k_0, that is, from k_0 the sequence $\{\tau_k\}$ is constant, being τ_{k_0}, that is, $\tau_{k+1} = \tau_k$ for

all $k \geq k_0$. By (28), we have that

$$F_{\tau_{k_0}}(\lambda_{k+1}) \leq F_{\tau_{k_0}}(\lambda_k) + F'_{\tau_{k_0}}(\lambda_k)(\lambda_{k+1} - \lambda_k) + \frac{L}{2}(\lambda_{k+1} - \lambda_k)^2. \tag{30}$$

Since $L = \sum_{i=1}^{n} \frac{b_i^2}{p_i}$, $F'_{\tau_k}(\lambda_k) = g_\phi(\lambda_k, \tau_k)$, and $\tau_k = \tau_{k_0}$ for all $k \geq k_0$, by (29) we have $F'_{\tau_{k_0}}(\lambda_k) = g_\phi(\lambda_k, \tau_{k_0}) = -L(\lambda_{k+1} - \lambda_k)$. Substituting the latter equality into (30), we obtain

$$F_{\tau_{k_0}}(\lambda_{k+1}) \leq F_{\tau_{k_0}}(\lambda_k) - \frac{L}{2}(\lambda_{k+1} - \lambda_k)^2, \quad \forall\, k \geq k_0.$$

Hence, and the fact that $\frac{\epsilon}{2} < |g_\phi(\lambda_k, \tau_k)| = L|\lambda_{k+1} - \lambda_k|$, we have

$$F_{\tau_{k_0}}(\lambda_{k+1}) \leq F_{\tau_{k_0}}(\lambda_k) - \frac{\epsilon^2}{8L},$$

that is, the sequence $\{F_{\tau_{k_0}}(\lambda_k)\}$ is decreasing for all $k \geq k_0$. ■

Theorem 4.2: *Suppose that for τ_{k_0}, with k_0 given in Lemma 4.1, the function $F_{\tau_{k_0}}$ is bounded from below by F^{low}. Then the method reaches the stopping criterion given in Step 2 in a number of iterations bounded from above by*

$$k_0 + 8L\left(F_{\tau_{k_0}}(\lambda_{k_0}) - F^{low}\right)\epsilon^{-2}. \tag{31}$$

Proof: By Lemma 4.1, we have that there exists k_0 such that if the method does not halt until iteration k_0, then

$$F_{\tau_{k_0}}(\lambda_{k+1}) \leq F_{\tau_{k_0}}(\lambda_k) - \frac{\epsilon^2}{8L}, \quad \forall\, k \geq k_0. \tag{32}$$

If $|g(\lambda_k)| \leq \epsilon$ for some $k < k_0$, then the method halts at a number of iterations less than k_0. Otherwise, by (32), we have

$$F_{\tau_{k_0}}(\lambda_{k+1}) \leq F_{\tau_{k_0}}(\lambda_k) - \frac{\epsilon^2}{8L}, \quad \forall\, k \geq k_0.$$

Let $k > k_0$ be an iteration in which the method did not halt. By Lemma (32), we have that

$$\begin{aligned}
F_{\tau_{k_0}}(\lambda_{k_0+1}) &\leq F_{\tau_{k_0}}(\lambda_{k_0}) - \frac{\epsilon^2}{8L} \\
F_{\tau_{k_0}}(\lambda_{k_0+2}) &\leq F_{\tau_{k_0}}(\lambda_{k_0+1}) - \frac{\epsilon^2}{8L} \\
&\vdots \\
F_{\tau_{k_0}}(\lambda_{k+1}) &\leq F_{\tau_{k_0}}(\lambda_k) - \frac{\epsilon^2}{8L},
\end{aligned}$$

whence it follows that

$$F_{\tau_{k_0}}(\lambda_{k+1}) \leq F_{\tau_{k_0}}(\lambda_{k_0}) - (k - k_0)\frac{\epsilon^2}{8L}.$$

Since F^{low} is a lower bound for $F_{\tau_{k_0}}$, for all $k \geq k_0$ we have

$$F^{low} \leq F_{\tau_{k_0}}(\lambda_{k+1}) \leq F_{\tau_{k_0}}(\lambda_{k_0}) - (k - k_0)\frac{\epsilon^2}{8L},$$

whence it follows that

$$k \leq k_0 + 8L\left(F_{\tau_{k_0}}(\lambda_{k_0}) - F^{low}\right)\epsilon^{-2}.$$

Thus we have proved that the number of iterations for which $|g(\lambda_k)| > \epsilon$ is finite. ■

In the optimization literature, first-order methods generally guarantee a complexity of order $O(\epsilon^{-2})$ with respect to the number of iterations required to obtain an approximate solution with a given tolerance, as can be seen in [18] when considering $p = 1$. By Lemma 4.1, if τ_0 is chosen so that $\nu(\tau_0) \leq \frac{\epsilon}{2C}$, then $k_0 = 0$. Hence, by (32), the number of iterations for the method to reach $|g(\lambda_k)| \leq \epsilon$ is at most $8L(F_{\tau_0}(\gamma_0) - F^{low})\epsilon^{-2}$. In this case, Algorithm SGTA has complexity of order $O(\epsilon^{-2})$ with respect to the number of iterations required to obtain $|g(\lambda_k)| \leq \epsilon$, thus achieving the same complexity as traditional first-order methods.

5. Numerical experiments

This section presents a detailed computational evaluation of the proposed Smoothing Gradient-Type Algorithm (SGTA) for solving the Continuous Quadratic Knapsack (CQK) problem. We first analyse the behaviour of the two smoothing choices used within SGTA, namely the Interior Point Smoothing (IPS) and the Neural Network Smoothing (NNS), under different instance structures. In a second step, we compare the quality of the solutions produced by SGTA with two state-of-the-art methods from the literature: the accelerated fixed-point algorithm (APF) from [1] and the Newton-type method from [11]. In this comparison, we focus on feasibility and accuracy, rather than CPU times, since highly optimized Newton-type implementations are known to be extremely fast.

All tests are performed on synthetic instances structured according to three classical patterns [1,11]:

- Uncorrelated: Parameters a_i, b_i, p_i independently and uniformly sampled from the interval $[10, 25]$.
- Weakly correlated: Parameters b_i uniformly drawn from [10,25], while a_i, p_i are uniformly sampled from $[b_i - 5, b_i + 5]$.
- Correlated: Parameters defined by the relationship $a_i = p_i = b_i + 5a_i$, with b_i uniformly sampled from [10,25].

Unless otherwise stated, we consider problem sizes $n \in \{10^3, 10^4, 10^5\}$. For each configuration, the reported values correspond to the average over ten independent instances.

5.1. Smoothing-based experiments: IPS versus NNS

In the first set of experiments, we evaluate the impact of the smoothing choice inside SGTA. We test three values of the smoothing parameter $\tau \in \{10^{-1}, 10^{-5}, 10^{-10}\}$ with fixed reduction factor $\beta = 10^{-15}$. The stopping criterion for these runs is

$$|g(\lambda)| < 10^{-7},$$

where g is the exact dual function defined in (13). For each instance type and problem size, we report:

- **Iter**: Number of iterations until convergence.
- **Time**: Execution time in seconds.
- **Res**: Final projected residual norm, quantifying the constraint satisfaction.

These attributes collectively capture performance (time, iterations), precision (residuals) and optimality.

In the uncorrelated instances (Table 1), IPS consistently converged across all tested configurations. For the largest smoothing parameter ($\tau = 10^{-1}$), IPS achieved rapid convergence with low residuals (below 10^{-7}). In contrast, NNS failed to converge, hitting the maximum iteration limit (1001 iterations) and leaving residuals significantly large (up to 5.56×10^6). As τ decreased to 10^{-5} and 10^{-10}, both algorithms converged successfully, exhibiting comparable performance. However, IPS consistently maintained a runtime advantage, being approximately two to three times faster than NNS, especially noticeable at size 10^5.

For weakly correlated instances (Table 2), both algorithms exhibited challenges at larger smoothing scales ($\tau = 10^{-1}$), with neither IPS nor NNS converging effectively, both hitting the maximum iteration limit and producing large residuals (in the order of 10^4 to 10^6). As the smoothing parameter reduced ($\tau = 10^{-5}$ and 10^{-10}), residuals remained high and neither method achieved the required tolerance. Despite non-convergence, IPS consistently demonstrated a lower computational runtime, approximately half that of NNS. The lack of convergence for both algorithms highlights that weak correlation significantly affects convergence behaviour negatively, necessitating either further tuning or smaller smoothing parameters.

Table 1. Uncorrelated instances ($\beta = 10^{-15}$).

		IPS			NNS		
τ	Size	*Iter*	Time	Res	*Iter*	Time	Res
10^{-1}	10^3	153	0.0013	9.15×10^{-8}	1001	0.0204	5.46×10^4
	10^4	160	0.0134	8.49×10^{-8}	1001	0.1913	5.52×10^5
	10^5	505	0.4416	6.20×10^{-8}	1001	2.4275	5.56×10^6
10^{-5}	10^3	116	0.0010	7.29×10^{-8}	114	0.0007	7.46×10^{-8}
	10^4	137	0.0116	9.76×10^{-8}	139	0.0302	6.92×10^{-8}
	10^5	404	0.3538	1.15×10^{-7}	746	1.9639	9.71×10^{-1}
10^{-10}	10^3	87	0.0007	6.98×10^{-8}	90	0.0005	9.78×10^{-8}
	10^4	108	0.0091	7.50×10^{-8}	91	0.0198	7.36×10^{-8}
	10^5	204	0.1821	1.07×10^{-7}	97	0.2517	1.11×10^{-7}

Table 2. Weakly correlated instances ($\beta = 10^{-15}$).

		IPS			NNS		
τ	Size	Iter	Time	Res	Iter	Time	Res
10^{-1}	10^3	1001	0.0083	1.75×10^4	1001	0.0098	1.31×10^4
	10^4	1001	0.0828	1.39×10^5	1001	0.2391	1.77×10^5
	10^5	1001	1.0211	1.69×10^6	1001	2.5954	2.84×10^6
10^{-5}	10^3	1001	0.0083	1.10×10^4	1001	0.0049	1.10×10^4
	10^4	1001	0.0837	1.36×10^5	1001	0.1820	1.36×10^5
	10^5	1001	1.0212	1.68×10^6	1001	2.3512	1.68×10^6
10^{-10}	10^3	1001	0.0083	1.34×10^4	1001	0.0048	1.34×10^4
	10^4	1001	0.0831	1.38×10^5	1001	0.1816	1.38×10^5
	10^5	1001	1.0219	1.63×10^6	1001	2.3495	1.63×10^6

Table 3. Correlated instances ($\beta = 10^{-15}$).

		IPS			NNS		
τ	Size	*Iter*	Time	Res	*Iter*	Time	Res
10^{-1}	10^3	79	0.0007	7.19×10^{-8}	1001	0.0205	5.46×10^4
	10^4	89	0.0075	7.73×10^{-8}	1001	0.2038	5.55×10^5
	10^5	457	0.3963	8.98×10^{-8}	1001	2.0343	5.56×10^6
10^{-5}	10^3	61	0.0005	6.73×10^{-8}	65	0.0004	5.60×10^{-8}
	10^4	73	0.0064	9.00×10^{-8}	353	0.0683	7.79×10^{-2}
	10^5	177	0.1555	1.58×10^{-8}	172	0.3948	1.66×10^{-1}
10^{-10}	10^3	46	0.0004	4.61×10^{-8}	46	0.0003	8.18×10^{-8}
	10^4	55	0.0048	5.94×10^{-8}	50	0.0094	8.34×10^{-8}
	10^5	64	0.0566	2.06×10^{-7}	54	0.1204	1.29×10^{-7}

With correlated instances (Table 3), IPS showed robust performance across all smoothing parameters, quickly converging with very small residuals (10^{-7} magnitude). At the coarse smoothing parameter ($\tau = 10^{-1}$), NNS failed to converge, reaching the iteration cap with extremely high residuals. However, with tighter smoothing parameters ($\tau = 10^{-5}$ and 10^{-10}), NNS successfully converged, although at times with noticeably larger residuals (up to 10^{-1} at $\tau = 10^{-5}$ and size 10^5). IPS consistently converged in fewer iterations and significantly less computational time compared to NNS, particularly noticeable for larger problem sizes.

The results of this first group demonstrate the superior robustness and efficiency of IPS relative to NNS across various instance types and smoothing parameters. IPS consistently converged rapidly and maintained small residuals, showing resilience even under challenging correlation structures. NNS, in contrast, exhibited sensitivity to larger smoothing parameters and correlation structures, often failing to converge at higher τ values and demanding significantly greater computational resources. This indicates that IPS is generally preferable due to its stability, lower runtime and consistent convergence, especially critical for large-scale CQK problems.

5.2. Comparison with APF and Newton-type method

In the second group of experiments we directly compare the quality of the solutions produced by SGTA with two reference methods from the literature:

Table 4. Residuals for SGTA (IPS/NNS), APF [1] and Newton-type method [11].

Type	Size	SGTA (IPS/NNS)	APF	NEW
Uncorrelated	10^3	0.00e + 00	2.04e − 10	1.16e − 10
Uncorrelated	10^4	1.05e − 09	2.10e − 09	4.19e − 09
Uncorrelated	10^5	7.45e − 10	2.10e − 07	1.90e − 07
Weakly corr.	10^4	1.70e + 05	1.52e+06	4.89e − 09
Weakly corr.	10^5	1.68e + 06	1.52e+07	1.37e + 05
Correlated	10^3	1.46e − 10	3.49e − 10	2.04e − 10
Correlated	10^4	0.00e + 00	5.59e − 09	3.26e − 09
Correlated	10^5	9.31e − 10	2.79e − 08	4.19e − 04

- the accelerated fixed-point algorithm (APF) from [1];
- the Newton-type method of Cominetti et al. [11].

All algorithms are run on the same instances and with a more stringent stopping criterion

$$|g(\lambda)| < 10^{-10},$$

to stress their accuracy. For SGTA, both IPS and NNS performed equally with a fixed smoothing setup

$$\beta = 10^{-30} \quad \text{and} \quad \tau = 10^{-20},$$

which ensures that the smoothing error is negligible at the scale of the desired tolerance. The APF and Newton codes are executed with the same data and stopping tolerance for a fair comparison.

Table 4 summarizes the final residuals $|g(\lambda)|$ obtained by each method.

For uncorrelated instances, SGTA attains residuals of the same order or smaller than those of both APF and the Newton-type method, including the largest dimension $n = 10^5$. This shows that, under an appropriate smoothing scale, the proposed algorithm matches the accuracy of highly tuned methods on this classical test family.

For correlated instances, SGTA again delivers very accurate solutions, with residuals around 10^{-10}–10^{-9}. In this setting, SGTA is systematically more accurate than APF, and substantially more accurate than the Newton-type implementation for $n = 10^5$, where the reported residual for Newton remains around 10^{-4}. These results indicate that the smoothing framework is not only viable but also competitive in terms of solution quality on structured correlated data.

For weakly correlated instances, all methods face difficulties, but with different profiles. SGTA and APF both stagnate with large residuals, revealing that the chosen fixed smoothing parameters (for SGTA) and the fixed-point dynamics (for APF) are not fully adequate for this adverse correlation pattern. The Newton-type method, by contrast, attains very small residuals for $n = 10^3$ and 10^4, but its performance deteriorates at $n = 10^5$ in this particular experiment. Overall, this family highlights a limitation of the current SGTA parameterization: weakly correlated data may require either a refined adaptive smoothing strategy or a hybridization with Newton-type steps to recover the same robustness observed in the other classes.

These numerical comparisons place SGTA in a clear position with respect to existing approaches. The semismooth Newton method [11] remains the most aggressive structure-exploiting solver, while APF [1] offers a very simple fixed-point alternative. The proposed SGTA, built on smooth approximations of the dual equation, attains solution qualities comparable to or better than these methods on uncorrelated and correlated instances, with a transparent convergence theory and without relying on intricate breakpoint logic. At the same time, the weakly correlated tests provide a benchmark to discuss the limitations of the current smoothing scheme and motivate future refinements.

6. Final remarks

In this paper, we introduced a Smoothing Newton-Type method aimed at efficiently solving the continuous convex separable knapsack problem. Our proposed approach simplifies existing techniques by replacing direct derivative computations with a smooth approximation of the dual function. We provided theoretical results confirming the algorithm's convergence and robustness under various conditions. Numerical experiments performed on synthetic problem instances reinforced the practical applicability and flexibility of our approach. The results demonstrated that our method effectively handles a variety of scenarios and scales, making it a promising option for solving large-scale continuous quadratic knapsack problems. To point out possible future developments, we note that there are many important problems that are modelled as quadratic problems where the matrices, even though positive definite, are not diagonal, thus preventing us from directly using the approach proposed in this work. Despite this, for illustrative purposes, we have observed the solution of SVM training problems, the combination of one-dimensional schemes with spectral projected gradient (SPG) in [11] and diagonal approximations in [19].

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